



A data-driven learning method for online prediction of drone battery discharge

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ARTICLE INFO

Article history:

Received 19 May 2022

Received in revised form 29 August 2022

Accepted 27 September 2022

Available online 3 October 2022

Communicated by Christian Circi

Keywords:

Battery state of charge

Drone

Segmentation

Trajectory prediction

UAS

UTM

ABSTRACT

This paper describes an adaptive method to predict the battery discharge of a multirotor drone over a generic path. A proper assessment of battery state of discharge trend is critical to ensure a safe operation of battery-powered aerial vehicles in critical environments, such as the urban ones. Several standard paths were executed by a commercial, battery-powered drone to acquire the data needed to train a Deep Learning based method; telemetry files and ground-collected data were processed to train the proposed method according to a trajectory segmentation strategy. Two learning configurations were trained to predict the time-of-flight and the integral of the battery current needed to fly the standard path segments. The current integral for each standard path segment is exploited to estimate the corresponding reduction of the battery state of charge. Based on path segmentation into predefined standard sections, the presented solution allows to predict time of flight and battery consumption along any generic path. This can be exploited to plan a complete path at strategic stage, as well as to estimate the remaining available power resource at any intermediate point along a generic trajectory both at strategic stage and during mission execution to be exploited by the drone operator or by a traffic management service. To validate the technique, a scaled package delivery mission is presented as an example of contingency management application. The maximum distance that the drone can reach from selected points along the mission path was computed according to the remaining battery level. Considering a random distribution of safe-landing areas around the planned path, the computed maximum distance was used to determine which safe-landing areas can be reached by the drone from a generic point of the path after an unexpected event.

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1. Introduction

The current worldwide interest in the development of an Unmanned Traffic Management (UTM) system is supported by the standardization of several procedures to allow safe operations of Unmanned Aircraft Systems (UASs). Several projects have been focused on the identification of services that must be provided to support UTM, such as flight planning, tracking, conflict detection and resolution, contingency management [1–8]. The so-called Trajectory Based Operations (TBO) and specific methods that have been introduced for Air Traffic Management (ATM) [9–12] can also be analyzed to identify solutions to speed up and manage the traffic flow of several UASs. Specifically, trajectory prediction is a

critical element for UTM to support safe and on-time operations [13]. Considering the developed Concept of Operations for UASs management [14,15] and according to current airspace regulations, specific UAS missions require the flight plan submission to the proper authority. The improvement of trajectory prediction is useful to better manage the increasing number of UAS missions during the strategic phase, i.e. when flight plans are submitted. This capability is also useful in the tactical phase, i.e. when several UASs are performing their missions in the same airspace volume. The prediction of specific critical parameters related to the trajectory execution [16], such as time-of-flight or battery/fuel consumption, can definitely contribute to support the UTM flight planning and management tasks [17]. The current Regulation about UAS management is experiencing several updates in order to face the increasing UAS market demand and the different profiles of UAS operations [14,18,19]. The development of standard scenarios [20] allows to define commonly used mission parameters for safe operations.

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This paper describes a Deep learning-based method to predict the UAS battery consumption over an assigned path for also evaluating the wind vector effects. The prediction of battery consumption allows to better develop the mission profile for a single UAS and to better manage the overall traffic flow in a specific airspace volume.

According to worldwide regulations, the UAS operator is responsible for the operation at any time. He must develop specific operative and contingency procedures according to the UAS model and mission, informing the proper authority and the involved people about the planned procedures. Considering battery consumption, during the strategic phase the UAS operator must compute the battery level required by the mission and he must monitor the remaining battery level to perform proper actions if it is too low and does not allow to safely complete the mission. The roles of UAS operator and remote pilot are stated by regulations, e.g. in [14,15]. Specific UTM services [1] shall be developed to support the UAS operator during any mission phase and several studies are focusing on the identification of proper systems to implement the identified services. A solution to predict the battery consumption can support the UAS operator during the strategic phase: for example, the operator of a multirotor UAS can interrogate the battery consumption predictor to evaluate the number of batteries needed to complete the planned route or to estimate the safe level of remaining battery to land also considering the wind effects. Moreover, the battery consumption must be monitored also during the tactical phase to perform proper actions if unexpected conditions occur. If the mission must be interrupted, the UAS can land immediately or fly to a safer area before landing. In this latter condition, the UTM system or the operator can re-plan the route also considering the predicted battery consumption.

The proposed method was developed and validated using a micro Remotely Piloted Aircraft Vehicle (RPAS) in automatic mode, so the word in common use *drone* is adopted. The paper is organized as follows: Section 2 describes some related works and projects, Section 3 reports the methodology and equipment used to develop the time-of-flight and the battery consumption predictors, Section 4 presents the results obtained by the proposed learning-based method, Section 5 describes an application of the proposed method, and at the end Section 6 includes the conclusions.

2. Related works

The recent increase of unmanned aerial traffic involves a challenging development of proper planning strategies to manage a high number of platforms, also considering the different levels of system automation [21]. A proper assessment of battery discharge level is a critical task especially for multirotor drones that are often characterized by limited autonomy performance, i.e. about 20-30 minutes for most commercial quadrotors, and strongly related to the environmental conditions. Several studies are interested in the prediction of the characteristic battery parameters, such as State Of Charge (SOC), State Of Health (SOH), or Remaining Useful Life (RUL), and different methods have been developed [22–25].

The study described in [26] is related to the prediction of an electric aircraft battery discharge. The battery modelling is defined, and the battery state is estimated by using an approach based on Unscented Kalman Filter algorithm. The battery state estimate and the future loads are used to predict the battery demand. The study reported in [27] proposes a method to predict the battery RUL of an electric vehicle thanks to the battery SOH estimation and exploiting real-word data. In [28] the authors use an improved particle filter and a sliding-window grey model to predict the battery RUL.

The study reported in [29] compares four Machine-Learning-based methods, i.e. Least Absolute Shrinkage and Selection, Least

Squares Support Vector machines for Regression, Multi-Layer Perceptron, and Gradient Boosted Trees, to predict the battery RUL adopting an hexa-copter platform. In [30] the authors adopt a Long-Short-Term Memory (LSTM) method to predict the battery SOC, a multiple-constraints method to estimate the battery State Of Power (SOP), and then a joint estimation applying a Kalman filter for SOC and SOP estimation. The authors in [31] use a LSTM algorithm for sensor output prediction, then the faults are classified by using a Deep Belief Network and at the end the battery RUL is estimated.

This paper proposes a data-driven approach aimed at predicting the drone time of flight and the absorbed current integral over a generic path. Thanks to the data-driven approach, the presented approach does not require the knowledge of the drone dynamics and new features can be added varying the training dataset. The inputs are related to the planned path geometry and the wind condition, so this strategy was developed to fit the requirements of the UTM scenario. Moreover, the prediction of the absorbed current integral can be directly used to compute the battery SOC if the battery capacity is known. Otherwise, a method to estimate the battery capacity is reported in Appendix A.

3. Proposed methodology and equipment

The nominal maximum time-of-flight for a specific drone model is usually reported in the flight manual and it is related to the nominal battery capacity for micro and small drones. For most of commercial drones the pilot terminal reports the battery consumption as the remaining battery level, i.e. the SOC, as percentage of the battery capacity and this value is displayed on the controller and saved in the telemetry file. The battery level is usually reported as integer percentage.

The evolution of the battery SOC during a flight depends instantaneously on the discharge current, i.e. the overall current absorbed on board, and on the battery capacity that is a specific parameter for each battery type and, of course, drone model. The proposed method aims at predicting the integral of the discharge current for trajectory segments, and then computing the corresponding variation of battery SOC thanks to the knowledge of the battery capacity. It must be taken into account that such a prediction is challenging because the battery discharge is characterized by a non-linear behaviour and is influenced by several parameters, such as speed and manoeuvres, trajectory geometry, wind speed and orientation, payload operation, battery life cycle. By referring the prediction to the current integral for standard path segments rather than to battery SOC, the implemented predictor is valid for the specific drone and the considered flight conditions (speed, mass at take-off, etc.) for any capacity of the battery in use.

The battery SOC is mostly used to monitor flight data because it gives an immediate warning if the battery discharge is approaching a critical value. The battery SOC is related to the integral of the absorbed current and it can be computed with the Coulomb Counting method [25,30]:

$$SOC_t = SOC_{t_0} - \frac{1}{Q} \int_{t_0}^t c(\tau) d\tau \quad (1)$$

where SOC_t is the battery SOC at the time instant t , SOC_{t_0} is the battery SOC at the time instant t_0 , Q is the battery capacity, and $c(\tau)$ is the discharge current.

Besides the battery level, the remaining flight time is of great interest because it directly influences the mission management. This parameter is often reported on the pilot display of commercial drones. It directly depends on the remaining battery level, i.e. the

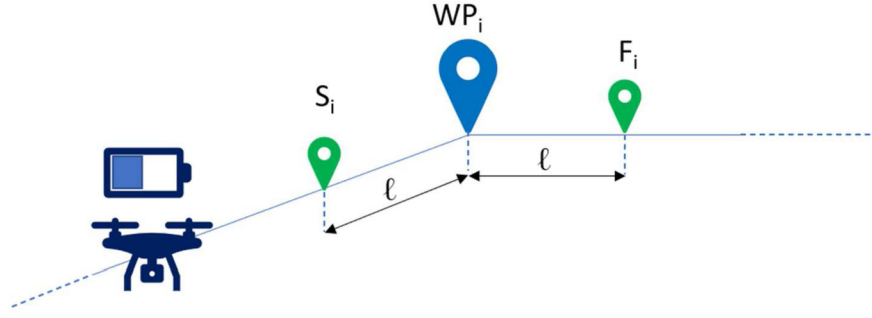


Fig. 1. Configuration of a standard segment. For the i^{th} corner S_i is the corner starting point and F_i is the corner ending point.

instantaneous SOC, but its derivation is complex because it needs the prediction of the battery discharge. So, the prediction of the remaining available flight time is a challenging task, but it is strongly needed.

3.1. Data acquisition

The proposed study adopts a learning-based method to predict the battery SOC thanks to the trajectory segmentation approach described by the authors in [32] to predict the drone time-of-flight for collision detection applications. The trajectory segmentation approach requires first to generate the planned path as a sequence of 3D waypoints and then to cut the planned path around the waypoints in sub-paths that are called standard segments or corners. The generic i^{th} corner is centred in the i^{th} waypoint WP_i and includes two legs: the initial leg from the corner starting point S_i to the waypoint WP_i and the ending leg from the waypoint WP_i to the corner ending point F_i . The points S_i and F_i are the intersecting points between the circle of assigned radius ℓ and the planned path, so that the starting and ending legs would have the same length, as reported in the Fig. 1. The length ℓ is fixed on the basis of the drone manoeuvring characteristics [32].

The proposed study was conducted adopting the DJI® Mavic 2 Enterprise Dual [33] commercial drone and the Ground Software Station UgCS® [34] to upload and manage pre-planned flights. The standard DJI® battery of the Mavic 2 series has 3850 mAh nominal capacity. The portable weather meter Kestrel 5500® [35] manufactured by Weather Instruments was used to acquire weather data on ground. The weather meter was fixed on a tripod with a rotating vane mount to align the weathervane to the wind direction. Considering the available weather data, the wind direction and intensity data are needed for the proposed method. Specifically, the weather meter allows to compute the intensity of head and cross wind components and the pointing direction with respect to the magnetic North. As a result, the wind vector $\vec{w}$ was computed.

The drone performed a properly selected number of laps in terms of *standard paths* to acquire the data for the learning method training and *generic paths* to test the developed method. A *standard path* is a closed route planned with a characteristic turning angle -such as equilateral triangle, square, hexagon- and with the same-length sides of 30 m so that they consist of a series of standard path segment, i.e. a sequence of corners. On the other hand, a *generic path* is a closed route planned with different turning angles and side lengths. Both *standard paths* and *generic paths* were executed at constant altitude Above Ground Level (AGL) of 20 m and at constant ground speed of 5 m/s. The ground speed of 5 m/s was selected to use a nominal value that is adopted in standard scenarios according to [20]. The flights were executed in automatic mode uploading a pre-planned route and setting a turn and stop manoeuvre when the drone reaches the WP_i to guarantee a soft turn for the drone payload.

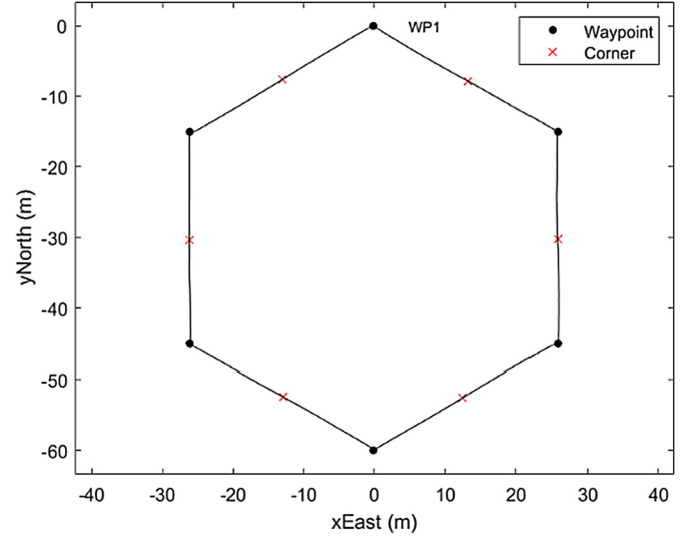


Fig. 2. One lap drone trajectory over the *Standard Path* with 120 deg characteristic angle. East North Up reference frame centred at first WP.

The considered *Standard paths* have a side length of 30 m, and the characteristic angle is a multiple of 30 deg. So, the following characteristic angles were considered: 0 deg - star shaped geometry, 30 deg - sawtooth shaped geometry, 60 deg - equilateral triangular geometry, 90 deg - square geometry, 120 deg - regular hexagon geometry as reported in the Fig. 2, 150 deg - regular dodecagon geometry, and 180 deg - straight line geometry. Each *standard path* was performed several times to acquire the needed data.

3.2. Data processing and clean

To properly implement the learning method, training, validation, and test datasets must be built from experimental data. Drone telemetry data, acquired according to the drone telemetry rate, must be processed. The proposed supervised learning approach requires to identify the telemetry data that are associated to each performed path corner. According to the drone telemetry rate, the corner limiting points S_{ti} and F_{ti} were identified as the telemetry samples with the minimum Euclidean distance to the planned S_i and F_i points, respectively. The telemetry samples included in the range that is limited by S_{ti} and F_{ti} belong to the considered path corner. Each flown corner represents a sample that will be used to train the learning method.

For the generic i^{th} corner the following six parameters were considered:

1. the corner characteristic angle, i.e. the inner angle between the corner legs,

Table 1
Characteristics of the NN architectures.

Parameter	Value
Training Function	Bayesian Regularization Backpropagation
Network Architecture	Feedforward, fully connected
Training, Validation, and Test Datasets	85%, 0%, 15%
Number of Epochs	1000
Number of input data	4
Number of output data	1

2. the angle between $\overline{w}$ and $\overline{S_i W P_i}$,
3. the angle between $\overline{w}$ and $\overline{W P_i F_i}$,
4. the wind intensity $|\overline{w}|$,
5. the time to fly from S_{ti} to F_{ti} ,
6. the integral of the absorbed current over the time interval to fly from S_{ti} to F_{ti} .

The first input is the nominal inner corner angle. The second and the third inputs are computed considering the nominal course angle and the measured wind direction. The wind intensity input is measured by the weather station. The time to fly and the current integral are computed by using the telemetry data.

The angles between the wind direction and the two legs of the corner were approximated to the nearest multiple of 30 deg angle. Wind intensity was reported in m/s in the telemetry files. The correspondent Beaufort category [36] was associated to classify the wind intensity in a limited number of categories. The performed flights include wind intensity values from the category 0 to the category 3, i.e. from calm wind up to 10 kts, that are wind conditions in which most drone operations are allowed.

The current integral included in (1) can be numerically computed over the i^{th} corner flight-time interval on the basis of telemetry data by using the trapezoidal rule [37] and it is named $curr_i$ in (2)

$$curr_i = \sum_{k=1}^{K_i} \frac{c_{k-1} + c_k}{2} (t_k - t_{k-1}) \quad (2)$$

where c_k is the current value at the time instant t_k , c_{k-1} is the current value at the time instant t_{k-1} , and K_i is the number of the available telemetry samples for the generic i^{th} corner according to the telemetry data rate which is not rigorous. This value is used to evaluate, by means of equation (1), the battery SOC change associated to the flight over the considered corner.

The initial database includes the data of all the corners flown during several drone flights. The standard deviation was computed for the time of flight and for the integral of current for each kind of standard corner, and the presence of several out-data in the collected samples was noticed. This strongly affects the learning capability, so a threshold of 3σ was set to both time and integral of current datasets: samples outside the $\pm 3\sigma$ threshold with respect to the mean value are not used to train the Neural Network (NN). After out-data removal, 726 valid samples were included in the database on the whole.

4. Predictor development and results

The proposed method aims to predict the time-of-flight and the integral of the battery discharge current over a path corner. So, proper learning algorithms must be identified.

4.1. Learning phase

After analyzing different Deep Learning algorithms [38–41] and realizing preliminary tests [42], the feedforward NN architecture was selected. The feedforward NN architecture was adopted as

classifier considering different path geometries that were executed by the drone in several wind conditions, according to the strategy described in [42]. Two independent feedforward NNs were developed, i.e. the Time-Predictor (TP) NN and the Battery-Discharge-Predictor (BDP) NN, to predict, respectively, the time-of-flight and the current integral over a standard path corner. Both NNs were trained by using a discrete range of path geometries -that are called *standard paths*- and categories of wind conditions. The MATLAB® Deep Learning Toolbox™ was adopted, and the Bayesian Regularization Backpropagation function was selected to train both NNs because it allows to achieve better results with large variability data [39,43,44]. Moreover, this function does not need a validation dataset. The samples were randomly assigned to the datasets. The main parameters that were set to develop both NN architectures are reported in Table 1.

To properly define the number of hidden layers and neurons, an incremental approach was followed analyzing the 1-hidden-layer and 2-hidden-layers architectures with increasing numbers of neurons. The NN performance of each considered configuration, i.e. numbers of layers and of neurons was estimated by using the Normalized Root Mean Square Error (NRMSE):

$$NRMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{v_t - v_p}{v_t} \right)^2} \quad (3)$$

where v_t is the true value of the predicted parameter, i.e. flight-time or current integral, v_p is the predicted value, and N is the number of samples, i.e. the number of samples in the training or test dataset. To reduce the dependency of the NN performance on the sample assignment to training and test datasets, the training phase was repeated 10 times with the random assignment mode for each NN configuration which was considered; the mean value of the NRMSE over the 10 repetition was used as figure of merit for each considered configuration. The increase of the neuron number was stopped when the NRMSE reaches an almost constant value. Following this strategy, a maximum number of 10 neurons in each layer was reached.

The 2-layer architecture was selected because it shows a better generalization behaviour. To select the optimal number of neurons in each hidden layer, the mentioned (average) NRMSE in percentage and the relative number of effective NN parameters that was computed thanks to the MATLAB® Deep Learning Toolbox™ routines in use [45] were analyzed. R is the ratio between the number of effective NN parameters and the total number of NN parameters in percentage. Fig. 3 shows the resulting performance of the 2-hidden-layer architecture, for the TP NN and for the BDP NN, varying the number of neurons. The displayed number of neurons is the total number of neurons in the two layers. The plot reports the percent values of NRMSE and R .

The number of neurons in the first and in the second hidden layers that allows a low NRMSE in the almost constant region of R results to be, respectively, 4 and 6 for the TP NN, whilst it is 5 and 6 for the BDP NN. Table 2 reports the performance of the selected NNs for training and test datasets.

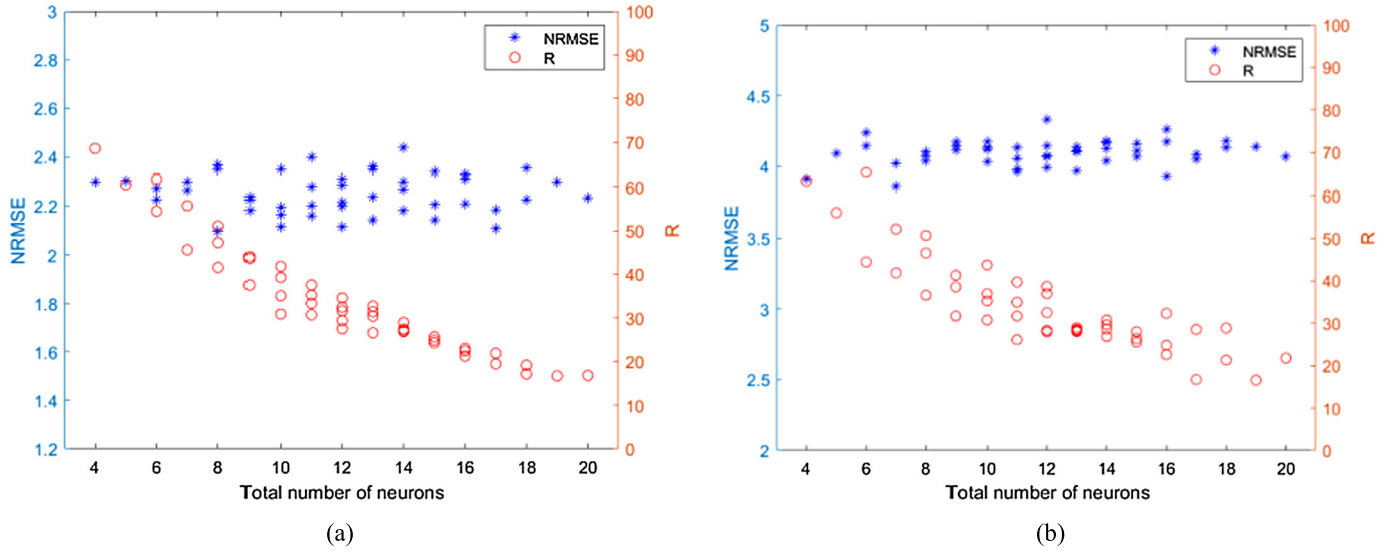


Fig. 3. NRMSE (left-axis) and R (right axis) for different 2-layers NN configurations varying the number of neurons. a) Time-Predictor (TP) NN performance. b) Battery-Discharge-Predictor (BDP) NN performance.

Table 2

NN performance for the selected Time-Predictor (TP) NN and Battery-Discharge-Predictor (BDP) NN.

NN (neurons in the two hidden layers)	NRMSE		R
	Training	Test	
Time-Predictor NN (4,6) neurons	2.16%	2.19%	39.3%
Battery-Discharge-Predictor NN (5,6) neurons	3.88%	3.98%	39.7%

4.2. Test phase

Four *generic paths* were designed and flown four times to test the method, as reported in Fig. 4. The selected paths mimic the route for a typical surveillance or last-mile packet delivery mission. To predict the time-of-flight and the integral of the current over *generic paths*, they were segmented in corners and the predicted values over each corner were summed; linear extrapolation over straight segments longer than the 180 deg standard corner was adopted according to the strategy reported in [32].

The results computed for the *generic paths* based on corner prediction are reported in the Table 3.

The sides of Path 1 and Path 2 have all the same length, and it is equal to a standard corner leg, so the predicted values of time and current for the complete path were computed summing the predicted values over corners. Instead, Path 3 and Path 4 involve different length sides, so the predicted values of time and current were computed summing the predicted values over corners and adding proportional values for the remaining part of sides. For the time prediction, the constant speed value of 5 m/s was considered, for the integral of current prediction a mean value of current integral was estimated with straight path flight tests.

Table 4 reports the true values and the predicted values of time and current integral over a single lap of each path. The predicted results are lower than the true values because the proposed learning method was trained collecting data in a discrete number of categories that involves specific path geometry and wind condition. So, the generalization capability is affected by the training dataset. However, the difference between the true values and the predicted values is within an acceptable range in terms of flight time and absorbed current integral that are critical parameters for most of current applications.

4.3. Battery SOC estimation

The battery SOC variation due to flying the generic i^{th} corner is

$$\Delta SOC_i = SOC_{i,in} - SOC_{i,fin} \quad (4)$$

where $SOC_{i,in}$ and $SOC_{i,fin}$ are the battery SOC at S_{ti} and at F_{ti} , respectively. It can be estimated by inverting equation (1) as follows:

$$\Delta SOC_i = \frac{1}{\bar{Q}} \int_{S_{ti}}^{F_{ti}} c(\tau) d\tau \quad (5)$$

in which the value of integral of the absorbed current is the one predicted by the developed BDP NN and $\bar{Q}$ is the properly estimated value of the battery capacity. Battery capacity is crucial in this computation, and it may be significantly different from the nominal value due to the battery history of charges and discharges, environmental conditions, etc. Appendix A reports a method to properly estimate the actual battery capacity, otherwise, the nominal battery capacity can be used.

Following the above procedure, the battery SOC was predicted over the *generic paths*. The following batteries were used for the *generic paths*:

- B01 for Path 1 and Path 2;
- B02 for Path 3;
- B03 for Path 4.

The overall travel of four rounds was considered as test for each path. The True Battery ΔSOC was computed on the basis of the telemetry data as the difference between the battery SOC at the first waypoint at the beginning of the first round and the battery SOC at the last waypoint of the fourth round of each path. The Predicted Battery ΔSOC was computed by using equation (5) with the predicted value of the current integral. The true and predicted values of battery ΔSOC and integral of current are reported in Table 5.

Due to the mentioned integer approximation of battery SOC in telemetry data, the proposed method of prediction is in substantial agreement with the reference data.

5. Battery-discharge-predictor application

The proposed method was tested to predict the battery ΔSOC during a generic planned path. Among the services required by the UTM framework, the Path Planner module can be supported by the proposed method to evaluate if the planned path can be executed by the drone guaranteeing a proper remaining battery SOC

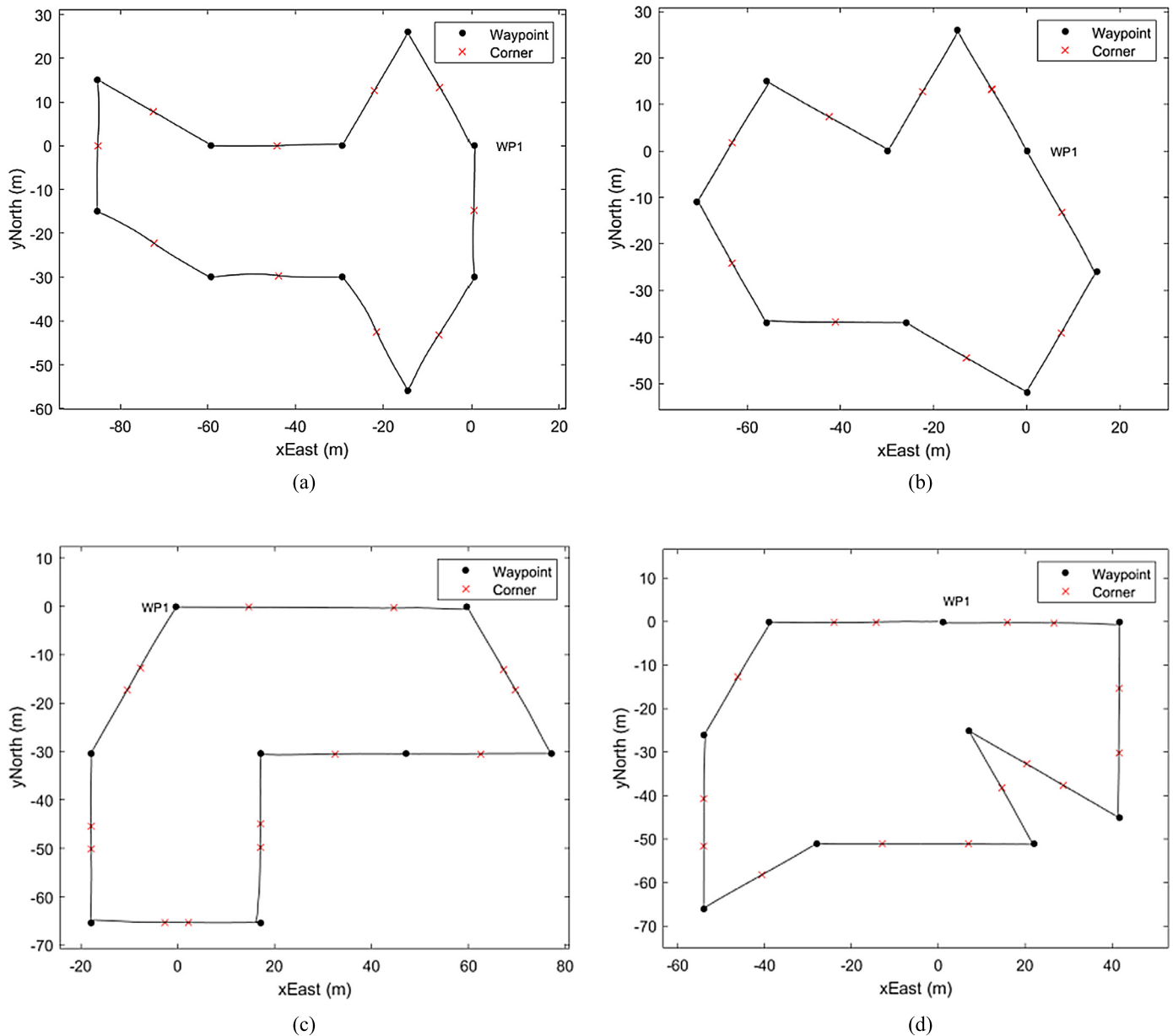


Fig. 4. One lap drone trajectory over *generic Paths*. East North Up reference frame centred at first WP. a) Path 1. b) Path 2. c) Path 3. d) Path 4.

Table 3

NN performance for Time-Predictor (TP) NN and Battery-Discharge-Predictor (BDP) NN over four *generic paths*. Each path was executed four times. The TP NN involves 4 and 6 neurons in each hidden layer. The BDP NN involves 5 and 6 neurons in each hidden layer.

NN (neurons in the two hidden layers)	NRMSE			
	Path 1	Path 2	Path 3	Path 4
Time-Predictor NN (4,6) neurons	9.61%	8.98%	10.20%	12.70%
Battery-Discharge-Predictor NN (5,6) neurons	13.79%	16.28%	13.94%	16.66%

Table 4

True value and predicted value of time and current integral over the first lap of each *generic path*.

	Time of flight (s)		Absorbed current integral (As)	
	True	NN Prediction	True	NN Prediction
Path 1	129.7	120.6	962.4	927.8
Path 2	112.3	108.8	884.6	839.8
Path 3	114.0	107.5	853.2	836.8
Path 4	130.1	124.9	984.6	967.5

Table 5

True and predicted values of the battery Δ SOC and of the integral of the absorbed current for the four *generic paths*. Each path was executed four times.

	Battery Δ SOC		Integral of absorbed current (As)	
	True value	Predicted value	True value	Predicted value
Path 1	32%	29.7%	3702.1	3630.3
Path 2	30%	26.7%	3482.8	3272.2
Path 3	27%	25.6%	3233.6	3212.1
Path 4	32%	31.2%	3823.5	3756.7

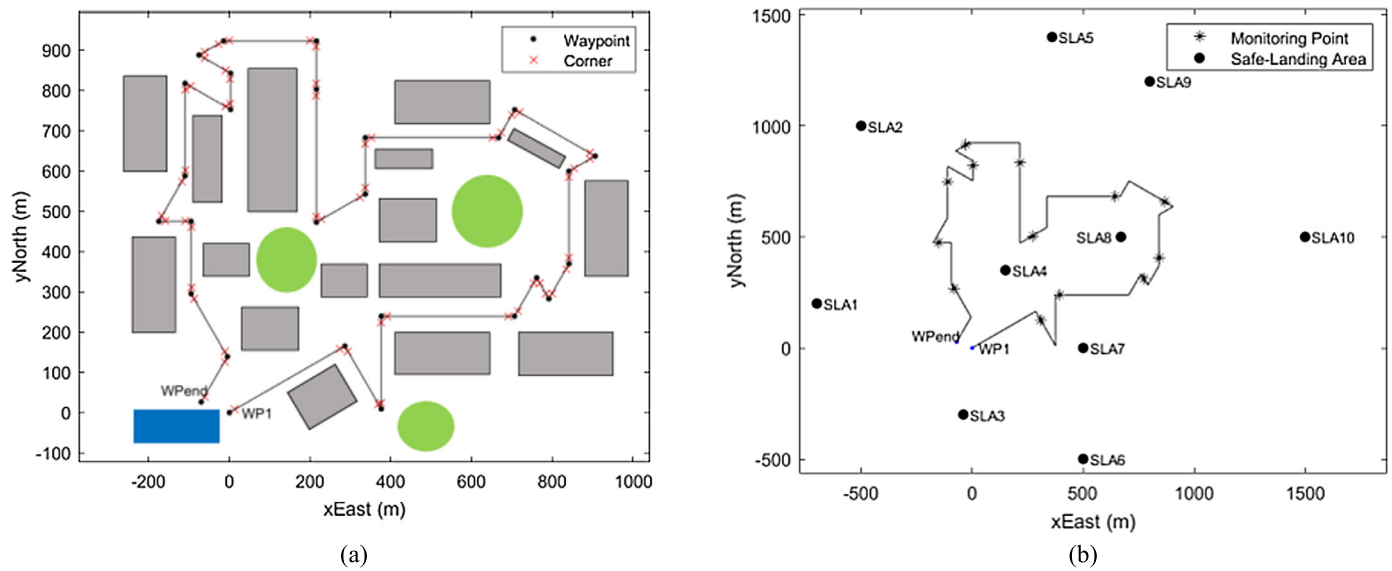


Fig. 5. Scaled package delivery path. a) The planned path is represented with black solid line, the package delivery warehouse with the blue rectangle, the fixed obstacles, such as buildings, with grey rectangles, and the SLAs with green circles. b) The planned path is represented with black solid line, the selected MPs with black stars, and the SLAs with black circles. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

to land or to face up unexpected situations. Moreover, the proposed method can also be applied to support tactical re-planning operations. At the beginning, a generic path was planned to simulate a scaled package delivery operation. Then, the battery SOC was predicted along the path and the maximum reachable distance was computed, considering airspace sectors that will be involved in most of typical drone operations, such as urban or industrial areas. This strategy allows to evaluate the area that the drone can cover with the remaining battery resource. An example of contingency management application is reported in the next paragraphs of this Section.

5.1. Battery SOC prediction over a planned path

A simulated scaled last-mile package delivery mission was planned. The drone starts the mission at the waypoint WP_1 and ends the mission at the waypoint WP_{end} following the waypoint sequence according to Fig. 5. The path was planned to simulate an urban environment in which the drone must reach planned waypoints to drop off a small package. Fig. 5a shows a schematic view of the planned mission. The grey rectangles are fixed obstacles, such as buildings, the blue rectangle is the warehouse for drone departure and arrival, and the green circles are the areas in which the drone could safely land. Wind conditions are supposed to be evaluated by using weather forecasts or in-site measurements at the time of the mission. The wind vector assumed for the simulation is along the direction 240 deg with intensity of 2.0 m/s, so the Beaufort category is 2. Fig. 5b shows the distribution of several Safe-Landing Areas (SLAs) that were defined with a random distribution in the mission surrounding area, and they are identified by black circles. Not all SLAs can be reached by the drone during the flight due to the limited remaining battery level. The battery SOC can be predicted in points belonging to the planned path during the strategic phase. The black stars in Fig. 5b show the points that were randomly selected for this application as Monitoring Points MPs. So, the MPs are the points in which the Predictors were applied.

Table 6 shows the predicted time-of-flight, current integral, and battery SOC at the selected monitoring points (MPs) as resulting from the flight along the planned path from WP_1 (mission star point).

Table 6

Predicted time-of-flight, current integral, and battery SOC from WP_1 (mission star point) at the selected monitoring points (MPs).

	Predicted Time of Flight (s)	Predicted Integral of Current (As)	Predicted Battery Δ SOC
MP1	76.0	609.6	95.0%
MP2	165.5	1322.7	89.2%
MP3	264.6	2114.3	82.7%
MP4	311.9	2487.7	79.7%
MP5	387.0	3084.3	74.8%
MP6	456.0	3632.3	70.3%
MP7	571.0	4553.5	62.8%
MP8	664.5	5300.3	56.7%
MP9	744.2	5935.2	51.5%
MP10	789.2	6290.6	48.6%
MP11	854.4	6808.5	44.4%
MP12	929.7	7407.2	39.5%
MP13	992.2	7906.1	35.4%

5.2. Determination of available safe-landing areas

In addition to the above discussed prediction of critical parameters for path planning and management, a strategy to support path re-planning is proposed here based on the available information resulting from the presented path prediction tool. The target is the tentative estimation of possible mission re-planning options along the nominal mission path, to be exploited in the case of contingencies preventing nominal completion of the mission. According to the environmental characteristics, two flight conditions can be adopted to reach a SLA:

1. Straight Path to safe landing. This condition can be exploited if the drone is flying at a safety height above obstacles before moving towards a given SLA.
2. Path to safe landing with corners. This condition can be exploited if the drone needs to avoid fixed obstacles to reach a given SLA performing corners.

First, the difference $\Delta SOC_{MP,th}$ between the battery SOC at each MP and a safety threshold of the SOC was computed; $\Delta SOC_{MP,th}$ represents the battery resources still available when reaching the considered MP which are available for the safe man-

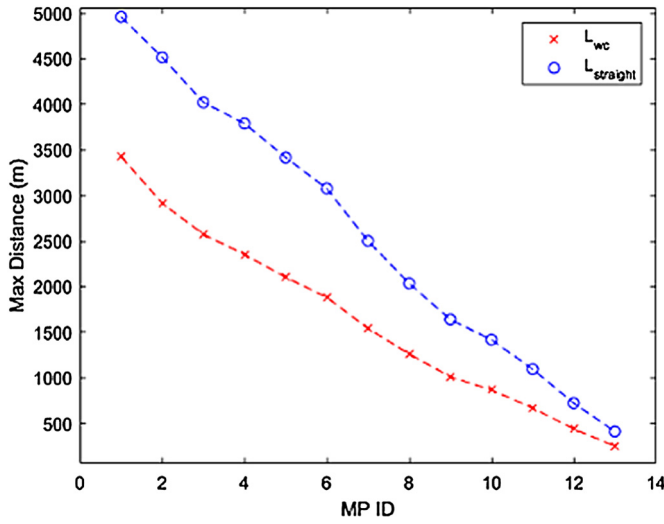


Fig. 6. $L_{straight}$ and L_{wc} distances at the selected path Monitoring Points.

agement of a contingency. In the following the threshold of 30% is adopted as a conservative threshold with respect to typical values [33,46,47] for safely landing.

About the Straight Path to safe landing condition, considering the constant-speed T180 *standard paths*, the corresponding mean value of battery SOC variation per unit flown distance in this *standard path* segment, i.e. in a straight direct path, $\Delta SOC_{ul, straight}$, was computed. So, the maximum distance, $L_{straight}$, that is expected the drone can fly from the considered MP over a straight-line path can be computed:

$$L_{straight} = \frac{\Delta SOC_{MP, th}}{\Delta SOC_{ul, straight}} \quad (6)$$

About the Path to safe landing with corners condition, the distance travelled from WP_1 along the planned path up to each MP was computed; also, the battery SOC decrease per unit of flown path ΔSOC_{ul} was derived as the variation of battery SOC divided by the flown distance; for a worst-case approach, 70% of the computed flown distance was considered in the computation of $\Delta SOC_{ul, wc}$. It represents a conservative estimate of the battery consumption being realized during the mission, influenced by (and somehow reporting) the specific characteristics of the flown mission path up to the considered path point. So, the maximum distance, L_{wc} , that is expected the drone can fly from the considered MP following a path with corners similar to the already travelled path can be computed:

$$L_{wc} = \frac{\Delta SOC_{MP, th}}{\Delta SOC_{ul, wc}} \quad (7)$$

Fig. 6 describes the difference of the computed maximum distances for the simulated test mission. Considering the same MP, Fig. 6 shows that L_{wc} is always smaller, even if slightly, than $L_{straight}$, as it is expected since in the first case the drone performs stop-and-turn manoeuvres moving towards the SLA.

So, from the knowledge of $L_{straight}$ and L_{wc} , it is possible to determine in the strategic phase if a SLA can be reached by the drone while over the planned path, as reported in Table 7. Two circles centred at each MP and with radius of $L_{straight}$ and L_{wc} were considered. The area of each circle describes the area that the drone can reach with the remaining battery level in straight-line flight or along an irregular path, respectively. Therefore, it is possible to determine if the defined SLAs can be reached by the drone from each considered MP. This approach can be used during the strategic phase to identify the SLAs that can be reached from MPs if an

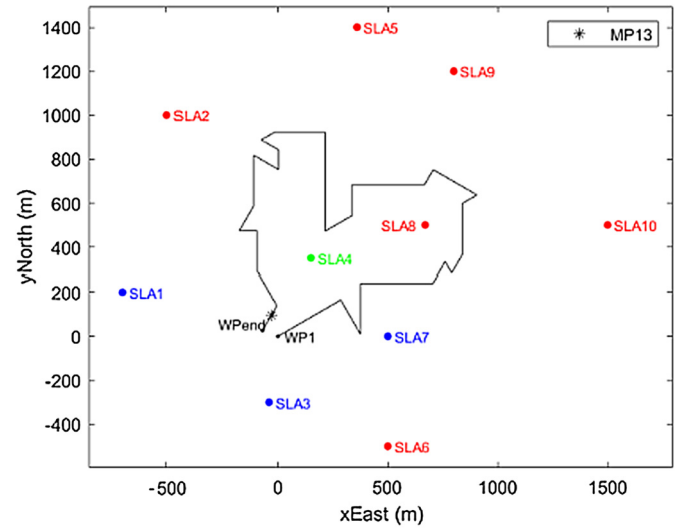


Fig. 7. Example of SLA availability determination in the strategic phase from the MP13 indicated with a black star. East North Up reference frame centred at first WP. The red SLAs cannot be reached by the drone, the green SLA can be reached by the drone according to both distances (6) and (7), the blue SLAs can be reached by the drone only according to the distance (6), so when the drone performs a straight-line path.

unexpected event does not allow to complete the planned mission. The concept is similar to the Point of No Return (PNR) for aircraft route planning. Table 7 shows the available SLAs for each MP considering $L_{straight}$ and L_{wc} .

Fig. 7 shows an example of SLA determination in the strategic phase. An unexpected event occurs when the drone reaches the MP13. The drone cannot land at the warehouse due to traffic bottleneck or area maintenance. For MP13, $L_{straight}=410.0$ m and $L_{wc}=248.9$ m. So, Fig. 7 reports the SLAs that cannot be reached (red), the SLA that can be reached with a straight-line path or with a more complex path (green) and the SLAs that can only be reached following a straight-line path (blue).

6. Conclusion

This paper describes a data-driven learning based method to predict the battery discharge of a multirotor drone over a generic planned path. Flight tests were realized to acquire huge amount of data that was processed according to a segmentation strategy in order to realize a proper database. The valid samples were used to train an adaptive neural algorithm based on the Bayesian Regularization Backpropagation approach. A Time-Predictor and a Battery-Discharge-Predictor were developed identifying the optimal neural network configuration for time-of-flight and integral-of-current prediction. The battery capacity was estimated, and the battery state of charge can be predicted starting from the current integral knowledge. The method was applied to a scaled package delivery mission that was planned considering a generic planned path and a random distribution of safe-landing areas in which the drone can land if an unexpected event occurs during the mission execution. The maximum distance that the drone can reach from selected points of the planned path was computed according to the remaining battery level and considering a safety threshold of the 30%. So, the safe-landing areas available for the drone can be identified.

The presented method was developed for a specific commercial drone, so the algorithm must be trained with a new dataset if a new drone model is involved. Moreover, the battery capacity must be estimated for each used battery and/or if the battery performance changes. To obtain an accurate estimation of both

Table 7

Available SLAs from each MP in straight-line flight (6) and irregular flight path (7). “Y” symbol indicates that the SLA can be reached by the drone from the corresponding MP. Bold “N” symbol indicates that the remaining battery level does not allow to reach the SLA.

	Reachable SLAs from selected MPs (straight line flight/irregular flight path)									
	SLA1	SLA2	SLA3	SLA4	SLA5	SLA6	SLA7	SLA8	SLA9	SLA10
MP1	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP2	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP3	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP4	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP5	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP6	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP7	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y
MP8	Y/Y	Y/Y	Y/Y	Y/Y	Y/Y	Y/N	Y/Y	Y/Y	Y/Y	Y/N
MP9	Y/Y	Y/Y	Y/N	Y/Y	Y/Y	Y/N	Y/N	Y/Y	Y/Y	Y/N
MP10	Y/N	Y/Y	Y/N	Y/Y	Y/Y	Y/N	Y/N	Y/Y	Y/N	Y/N
MP11	Y/N	Y/Y	Y/N	Y/Y	Y/N	N/N	Y/N	Y/N	Y/N	N/N
MP12	Y/N	Y/N	Y/N	Y/Y	N/N	N/N	Y/N	Y/N	N/N	N/N
MP13	Y/N	N/N	Y/N	Y/Y	N/N	N/N	Y/N	N/N	N/N	N/N

drone time of flight and absorbed current integral, the wind direction and intensity should be computed in the surrounding area of the flight. This method allows to support the path planning or re-planning tasks thanks to the development of safety procedures for Unmanned Traffic Management systems both at strategic and tactical levels; the latter case, in a real-time implementation of the algorithm, would benefit of the deterministic computational time guaranteed by the neural architecture of the proposed prediction tool.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

The estimation of battery capacity is a critical element to be computed. This study considers an experimental value of the battery capacity that was computed after tests data of the *standard paths* flights. Five LiPo DJI® Mavic 2 Intelligent Flight batteries were used. The batteries are identified as B01, B02, B03, B04, and B05.

Now, it is worth mentioning explicitly that each *standard path* was completed in more than one flight because a single battery was not sufficient to complete the path. For the generic complete j^{th} flight, following the Coulomb counting method and equation (1), the overall SOC variation can be related to the current integral:

$$SOC_{j,end} = SOC_{j,start} - \frac{1}{\tilde{Q}_j} \int_{t_{j,start}}^{t_{j,end}} c(\tau) d\tau \quad (\text{A.1})$$

where $SOC_{j,start}$ and $SOC_{j,end}$ are the battery SOC values at the beginning and at the end of the j^{th} flight and are available from the telemetry, $\tilde{Q}_j$ is the experimental determination of the battery being used in the considered flight, $t_{j,start}$ and $t_{j,end}$ are the time instant at the beginning and at the end of the j^{th} flight, respectively. The integral of the current in equation (A.1) can be numerically computed, following equation (2):

$$curr_j = \sum_{k=1}^{K_j} \frac{c_{k-1} + c_k}{2} (t_k - t_{k-1}) \quad (\text{A.2})$$

Table A.1

Experimental value of the capacity $\tilde{Q}_j$ for the batteries in use in the flights of the standard paths and mean value of the battery capacity Q_{mean} .

Battery ID	Standard Path Flight	$\tilde{Q}_j$ (As)	Q_{mean} (As)
B01	T30-1	12093	12234
	T90-2	12370	
	T120-1	12181	
	T180-2	12291	
B02	T0-2	12236	12522
	T30-2	12372	
	T30-3	12667	
	T60-1	12358	
	T150-1	12583	
	T180-3	12881	
	T180-4	12556	
B03	T0-1	12111	12025
	T0-4	11907	
	T60-2	11986	
	T90-1	12095	
B04	T0-3	12884	12699
	T150-2	12514	
B05	T120-2	12810	12944
	T180-1	13079	

where here K_j is the number of telemetry samples for the j^{th} flight, t_k and c_k are the relevant telemetry time and battery current samples. The value of battery capacity $\tilde{Q}_j$ was estimated from the telemetry data of the j^{th} flight as follows

$$\tilde{Q}_j = - \frac{curr_j}{SOC_{fin,j} - SOC_{in,j}} \quad (\text{A.3})$$

The Table A.1 shows the computed $\tilde{Q}_j$ value for the batteries in use in the flights of the standard paths, i.e. from B01 to B05, and the mean value of the battery capacity Q_{mean} .

The values reported in the Table A.1 were used to predict the battery ΔSOC in the tests of Sections 4 and 5.

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